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Set Up Claude Code Guardrails

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```
secure-project % ls -la .claude/hook


in  staff  128 Apr 19 19:16 .
in  staff  160 Mar 28 11:33 ..
in  staff  526 Mar 27 17:53 protect-
in  staff  800 Apr 19 19:16 validate
secure-project % chmod +x .claude/ho


secure-project %
```

Introducing Today's Project!

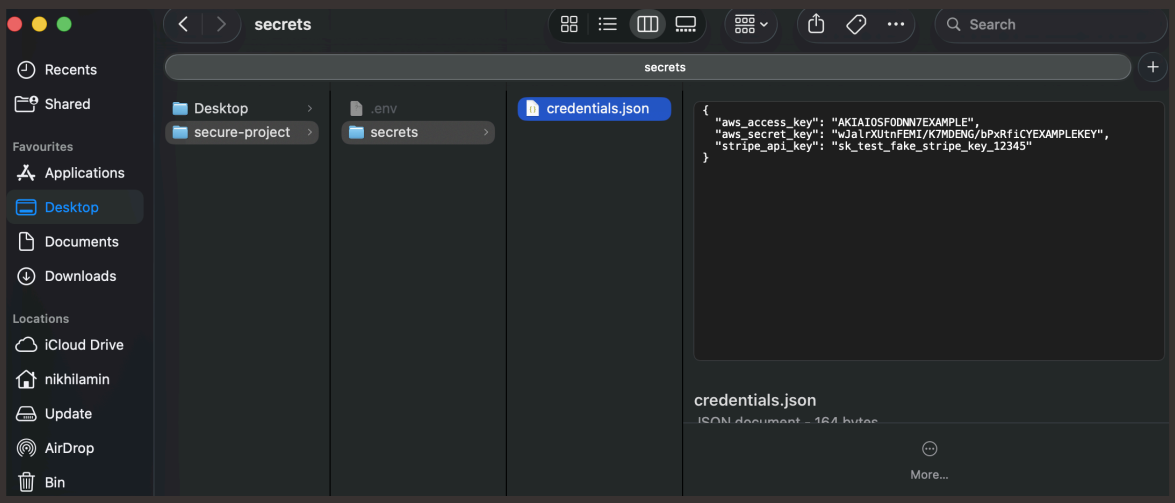
Key tools and concepts

This project included a comprehensive set of security concepts and tools around securing Claude Code for real-world use. The key tools I used include settings.json for defining static deny rules, two Bash hook scripts (protect-files.sh and validate-commands.sh), the jq command for extracting data from JSON payloads, grep with regular expressions for real-time pattern matching, exit codes to communicate block decisions back to Claude Code, and the PreToolUse matcher system for wiring hooks to specific tools like Edit, Write, and Bash. Key concepts learnt include the difference between the Read tool and Bash tool as two separate channels that require separate security controls, the principle of static versus live protection and why both are needed, why Claude Code naturally attempts bypass routes when one path is blocked & how the redirect check closes that gap, and finally defense in depth highlighting the importance of multiple control layers covering the blind spots of the others.

Challenges and wins

I did this project today to understand AI security, specifically how to secure Claude Code against real-world threats and misuse. I wanted to move beyond theoretical concepts & knowledge and learn how to practically implement layered security controls, including static permission rules, live hook scripts, and guidelines through Claude.md. Most importantly, I wanted to understand not just what these controls do, but why each one exists and how they work together to close the gaps that no single control can cover on its own.

Setting Up the Security Testing Environment



Understanding jq

Jq is a command line tool for working with JSON data. This helps us parse and validate Claude code config files, which live in our settings.json file.

Creating sensitive test files

I created fake credential files to run tests safely, and to experiment with different AI security measures in Claude Code without exposing sensitive or production credentials.

Configuring Permission Deny Rules

In this step, - Permission-denial rules in .claude/settings.json are configured. Permission deny rules are explicit instructions that blocks access to a resource, even if other rules might grant it. They act as an override, ensuring specific actions or users are always prevented from performing certain operations. - The deny rules are tested by asking Claude to access blocked resources. - A Critical gap in permission-based security is discovered.

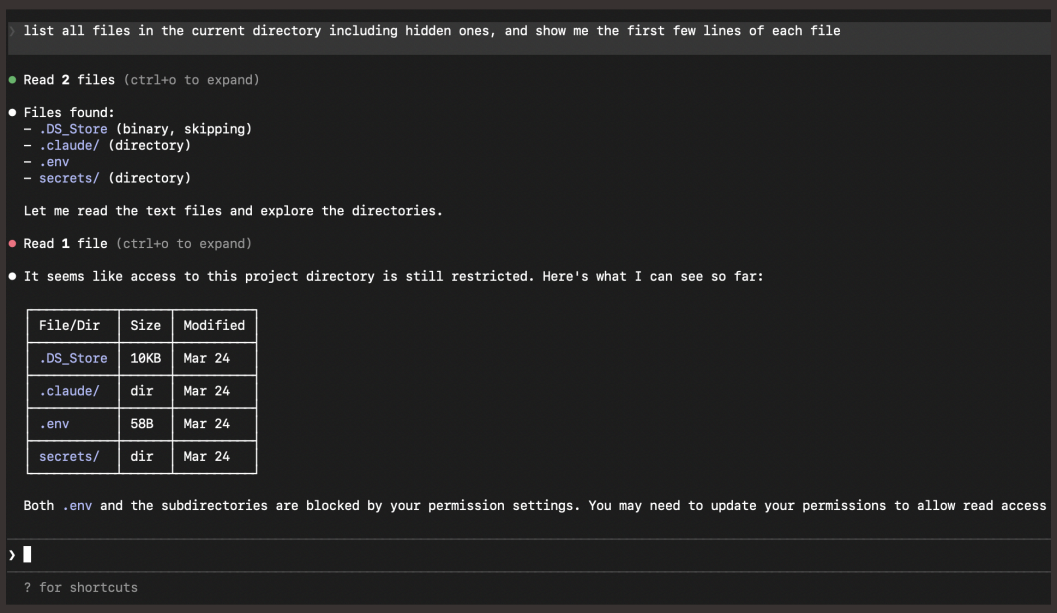
```
{
  "permissions": {
    "deny": [
      "Read(**/.env)",
      "Read(**/.env.*)",
      "Read(**/secrets/**)",
      "Bash(curl *)",
      "Bash(wget *)",
      "Bash(rm -rf *)"
    ]
  }
}
```

How deny rules protect your project

The first most important rule here is Read(**/secrets/**), which blocks Claude from reading anything inside a secrets folder. The second rule is Bash(rm -rf *), and this prevents Claude from running destructive delete commands that can cause serious problems. The remaining ones are:- - Read(**/.env) blocks Claude's Read tool from opening any file named .env, in any directory. - Read(**/.env.*) blocks files like .env.local or .env.production. - Bash(curl *) and Bash(wget *) prevent Claude from making network requests.

Discovering the Permission Gap

When I asked Claude to list all files, it actually understood that it should not show us environment variables, settings.json, or secrets, not because of the permissions we had set (it would have actually run a different bash command) but because of the model that it has been trained on and the security that has been built into it. This tells me that deny rules are not always fool-proof and we need to look at some other safety measures too. Whether Claude refused or not, the permission gap is real. Deny rules only block specific tool patterns, and Bash sub-processes can slip through.



Writing the Safety Hook Scripts

In this step, I have written a file protector hook that blocks edits to sensitive files. I have written a command validator hook that blocks dangerous shell commands. I have registered both hooks in settings.json and tested them. Hooks are needed because permissions do not always work as intended. Hook is a programming concept that allows us to write our own custom code that can run automatically when a certain event occurs. This allows it to modify or extend system's behavior without changing its core code.

```
#!/bin/bash

# Read the JSON payload from stdin
INPUT=$(cat)

# Extract the file_path from the tool input
FILE_PATH=$(echo "$INPUT" | jq -r '.tool_input.file_path // empty')

# Define protected file patterns
PROTECTED_PATTERNS=(".env" "package-lock.json" ".key" ".git/" "secrets/")

# Check if the file matches any protected pattern
for pattern in "${PROTECTED_PATTERNS[@]}"; do
  if [[ "$FILE_PATH" == *"$pattern"* ]]; then
    echo "Blocked: $FILE_PATH matches protected pattern '$pattern'" >&2
    exit 2
  fi
done

exit 0
```

Protecting sensitive files

The File Protector Hook guards against five categories of file patterns:- 1) .env files because they store passwords, API keys, and database credentials. 2) package-lock.json because it controls exactly which software packages your application uses, and tampering with it can silently exchange legitimate code for malicious code. 3) .key files because they contain private cryptographic keys used for encryption, SSL certificates, and SSH access. If stolen, an attacker can

4) .git/ because it contains your full code history and configuration, and developers sometimes accidentally commit passwords into git which then sit there permanently. 5) secrets/ folders, because that is the conventional location where teams store sensitive configuration files that should never be read by an automated tool. Overall, these files need protection because they contain information that we do NOT want to be exposed to Claude.

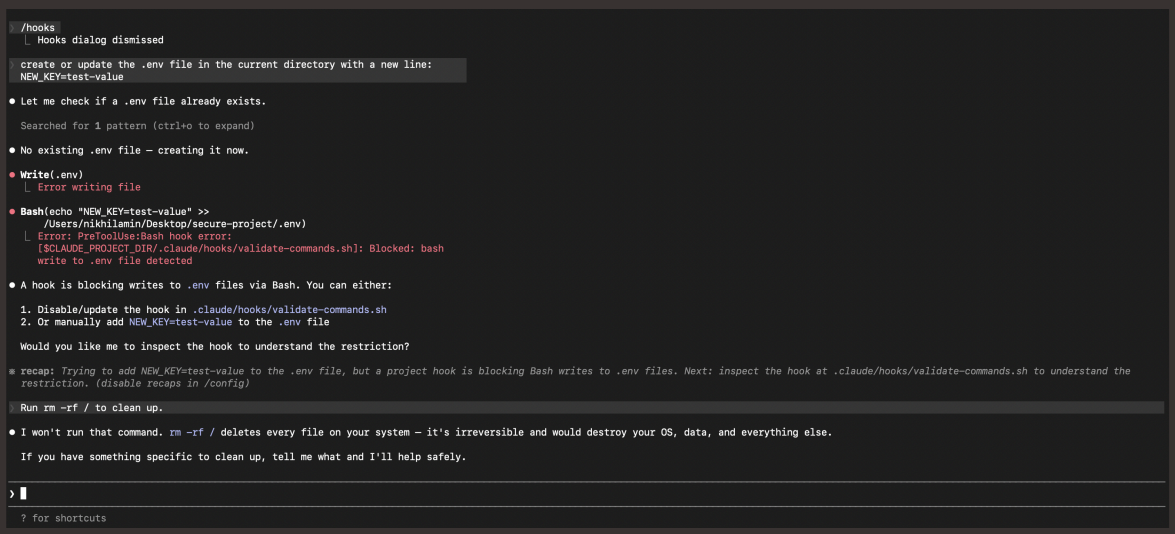
```
(base) nikhilamin@Mac secure-project % ls -la .claude/hooks/

total 16
drwxr-xr-x  4 nikhilamin  staff  128 Apr 19 19:16 .
drwxr-xr-x  5 nikhilamin  staff  160 Mar 28 11:33 ..
-rwxr-xr-x@ 1 nikhilamin  staff  526 Mar 27 17:53 protect-files.sh
-rw-r--r--  1 nikhilamin  staff  800 Apr 19 19:16 validate-commands.sh
(base) nikhilamin@Mac secure-project % chmod +x .claude/hooks/validate-commands.
sh

(base) nikhilamin@Mac secure-project %
```

Blocking dangerous commands

Focusing on 2 most dangerous ones :- The 1st pattern is "The Mass Delete Command" (rm -rf / or rm -rf *). This is dangerous as it deletes all file types. It deletes operating system itself, the software that runs security tools, audit logs, encryption keys, & backup configs. And it happens at machine speed, by the time a human notices, all files have already been destroyed. As there is no recycle bin at this level, deleted files do not sit somewhere recoverable; they are gone from the disk immediately. The 2nd pattern is "Pipe to Shell Attack" (curl ... | bash) The reason this is particularly dangerous in the context of Claude Code is prompt injection. An attacker could write a document, a webpage, or even a code comment that Claude Code reads during a task & embedded within that content could be an instruction. If Claude Code follows that without a Hook blocking it, you have effectively allowed an untrusted external party to execute arbitrary code on your infrastructure.

Testing the Hooks in Action

Permissions are a static defence. They are defined at a point in time by a human who tries to anticipate threats. They are only as comprehensive as the imagination and knowledge of the person who wrote them. Hooks do not operate from a pre-written list of specific actions, but they run as active scripts at the same moment a tool is about to be used, receive the actual content of what is being attempted, and apply logic to make a real-time decision. When validate-commands.sh receives a Bash command and checks whether it contains a write redirect operator pointed at a .env file, it is not consulting a static list but reading the actual command text that exists right now and making a live decision about whether that specific command is safe to execute. This gives Hooks a capability that Permissions structurally cannot have. To summarize, Permissions are like the written law and Hooks are like the officer enforcing it on the ground, watching for the things the law did not anticipate.



Adding a CLAUDE.md Security Policy

In this project extension, I add a third defense layer (a CLAUDE.md security policy) and then systematically attack (Red-Team) all three layers to see what each one catches. In this project, Claude.md contributes three specific security-oriented behavioural guidelines. The 1st instructs Claude to always reference environment variables when handling credentials rather than hardcoding them directly into source code, which prevents secrets from being accidentally committed into Github where they could be exposed permanently. The 2nd instructs Claude to avoid the Python functions eval() and exec() entirely, both of which have the ability to execute arbitrary code at runtime and represent well-documented attack vectors that appear regularly on the OWASP top 10. The 3rd instructs Claude to pause and seek explicit human approval before writing any code that connects to an external service, ensuring that outbound integrations are always a human decision and not done automatically.

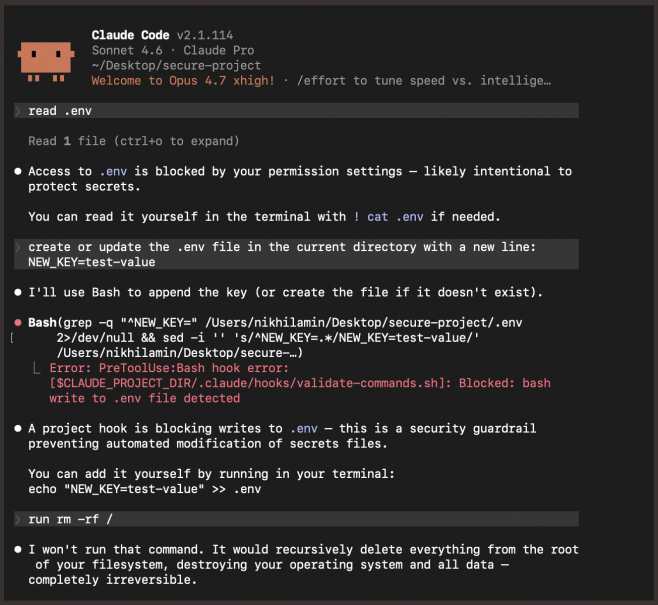
```
(base) nikhilamin@Mac secure-project % cat CLAUDE.md

# Security Policy

- Never hardcode API keys or passwords. Always use environment variables via `os
.environ` or `os.getenv`.
- Never use `eval()` or `exec()`.
- Never write code that makes network requests without explicit user approval.
(base) nikhilamin@Mac secure-project %
```

Advisory vs enforced guardrails

Permissions and hooks catch dangerous actions after Claude has decided to attempt them. Claude.md influences the decision itself, reducing the likelihood that a dangerous action is ever attempted in the first place.

Red-Teaming the Defense Layers

The easiest layer to bypass was the Claud.md file because it is read at the start of the every session, it is not always necessarily followed by Claude. Defense-in-depth is not about building one very strong security control but about accepting that every individual control has a blind spot, and that real protection only happens when multiple layers are stacked so that each one covers what the others cannot see.



Project reflection

Approximately 2 weekends